

Identification of studies via databases

Via citation searching

IDENTIFICATION

Records identified (databases): 2,436
Scopus 391 · IEEE 234 · ACM DL 1,811

After duplicates removed: 2,245

Duplicates removed: 191

Citation searching (snowballing)
backward + forward over the 54 studies

SCREENING

Automated pre-filter (Claude Opus 4.8)
2,245 title/abstract, retained 178

Excluded (pre-filter): 2,067
out of scope 1,877; other 190

Human dual screen: verification n = 385
178 retained + 207 (10% stratified)

Excluded at screening: 2,130
human-verified 270; auto-only 1,860

Retained for full text: 115
consensus: 87 include, 28 maybe

Pre-filter validation
 $\kappa = 0.79 [0.72, 0.85]$
 $FOR = 0.48\%$

Screened and assessed
against the same pre-registered criteria

ELIGIBILITY

Full-text assessed: 114 unique
1 duplicate found at full text

Excluded at full text: 60
no LLM oracle 55; other 5

Additional studies included: 29
same eligibility criteria applied

INCLUDED

INCLUDED STUDIES: 83

54 database search + 29 citation searching